

Subhrajit Prusty

SOFTWARE ENGINEER

Bengaluru, India

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Experience

Bolna

Bengaluru, India

LEAD BACKEND ENGINEER

January, 2026 - Present

- Building voice AI orchestration platform in Python, handling tens of thousands of concurrent calls across Ecommerce, EdTech, HealthTech, BFSI, and Hospitality
- Re-architected post-call processing into a phased, idempotent pipeline with atomic claim-and-checkpoint semantics, and hardened call-initiation paths against multi-pod race conditions — eliminated double-billing and lost-work bugs and enabled horizontal scaling without duplicate execution
- Cut per-pod worker memory by ~95% (from ~15GB to under 1GB) via WebSocket lifecycle cleanup; fixed graceful worker shutdown, recording-URL data loss, and webhook retry handling to absorb provider latency spikes
- Drove end-to-end observability with cross-service Datadog APM tracing keyed on a shared correlation ID across the async call lifecycle; stood up Loki + Grafana across all clusters with custom-metrics HPAs scaling on real application signals instead of CPU alone
- Architected the multi-cluster Kubernetes infrastructure (API, WebSocket, and self-hosted Deepgram inference clusters across regions) with namespace-oriented manifests, External Secrets Operator backed by AWS, and tuned ALB routing and resource limits for latency-sensitive WebSocket and SIP voice services
- Built **kubedump**, an open-source Go CLI that snapshots live Kubernetes resources into clean, Git-trackable manifests — deployed on a daily GitHub Actions schedule to give the team a continuous, reviewable history of cluster state without manual drift tracking
- Built a configurable LLM-based call analysis and disposition system for high-level per-call analytics, with model-gated execution and typed JSON configs to control cost and downstream correctness
- Hardened CI/CD with automated Alembic migrations and a backwards-compatibility check script enforcing rolling-deploy rules; stood up a dedicated webhook delivery worker with its own pipeline

skit.ai

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SOFTWARE ENGINEER - PRODUCT

November 2022 - August 2025

- Engineered an LLM-powered voicebot in Go, processing millions of calls for 40+ enterprise clients.
- Rebuilt Voicebot on the open-source Pipecat-AI framework, enabling modularity and third-party integration.
- Designed and implemented a bidirectional audio-streaming telephony stack with gRPC for real-time interactions.
- Configured and maintained in-house + OpenSIPS infrastructure for call routing with telephony vendors like TCN and Twilio.
- Designed and implemented a flow-configuration validation service, completely eliminating config-related incidents.
- Maintained and enhanced the voicebot dialogue management system, adding features and improving reliability.
- Optimized flow-configuration loading, reducing memory usage by 80%.
- Served as primary on-call engineer managing incident response for critical voicebot and telephony infrastructure.
- Maintained 99.5% uptime for voicebot systems using Grafana, Loki, OpenTelemetry, Prometheus for monitoring.

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SOFTWARE ENGINEER - SOLUTIONS

December 2020 - October 2022

- Owned and maintained a critical internal repository, establishing code review processes and authoring contribution guidelines; significantly improving code quality and reducing technical debt.
- Designed and optimized voice bot flow configurations in collaboration with cross-functional teams.
- Developed 3rd-party API integrations, including PCI-DSS compliant Payment APIs, improving voice bot's debt collection capabilities
- Enhanced build pipelines, reducing deployment time and improving CI/CD efficiency.
- Maintained, customized and enhanced internal ML models used for real time conversational voice bots.
- Designed and implemented an automated daily reports system that simplified accessing and transforming on-prem data

racetrack.ai (Internship)

Bengaluru, India

RESEARCH AND DEVELOPMENT TEAM

August 2019 - March 2020

- Using data collected by automated web-scraping, published an abstract on Personalized Hybrid-Recommender Systems using social network analysis.
- Using realtime image processing, developed a minor claims automation service for car accident processing, improving claim resolution efficiency.
- Maintained and improved the performance of the primary AI chatbot product, Marvin.

ITC Infotech, Innorruption Lab (Internship)

RESEARCH AND DEVELOPMENT TEAM

Bengaluru, India

May 2019 - July 2019

- Using image processing and automation, developed a metal sheet counting algorithm.
- Using facial detection and recognition, built an ID-card detail extraction system, optimizing concierge check-in process.

Education

Institute of Technical Education and Research (ITER)

B.TECH IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Bhubaneswar, Odisha, India

2016 - 2020

- CGPA : 8.81

Ispat English Medium School (IEMS)

INTERMEDIATE/+2 FROM ISC

Rourkela, Odisha, India

2014 - 2016

- Percentage : 84.5

Ispat English Medium School

MATRICULATION FROM ICSE

Rourkela, Odisha, India

2014

- Percentage : 94.17

Skills

Programming	Python, JAVA, C, Bash
Web	Apache, Flask, Selenium, Bootstrap, Wordpress, HTML5, CSS3
Technologies	AWS, GNS3, GitHub
Skills	Dev-ops, Linux System Administration, Web Development